Pharmaceutical Field Force Management Platform

Comprehensive Product Requirements Specification

Document Version: 1.0
Date: 19 August 2026
Product Type: Pharmaceutical Field Force Management / Sales Force Automation / Life Sciences CRM
Target Platforms: Android, iOS, Web Administration & Management Portal
Primary Users: Medical Representatives, Sales Representatives, Area/Regional Managers, Sales Managers, Medical Affairs/MSLs, Marketing, Commercial Operations, HR/Admin, Finance, Compliance, System Administrators
Geographic Assumption: Pakistan-first architecture with capability for multi-country deployment

---

1. Executive Summary

This product should **not** be designed as merely a GPS attendance application or a “doctor visit tracker.”

A serious pharmaceutical Field Force Management platform sits at the intersection of:

- Organizational hierarchy
- Employee management
- Territory management
- HCP/HCO master data
- Customer segmentation
- Targeting
- Cycle planning
- Tour planning
- Daily activity management
- Field visits
- Call reporting
- Product detailing
- Sample management
- Orders and sales activity
- Expenses
- Attendance and leave
- GPS/location intelligence
- Digital content
- Events
- Medical inquiries
- Adverse-event capture
- Performance management
- Manager coaching
- Analytics

- Notifications
- Offline mobile operation
- Compliance
- Auditability
- Integration with ERP, HR, distributor, MDM and analytics systems
- Increasingly, AI-assisted planning and next-best actions

This conclusion is consistent across the major life-sciences CRM platforms.

Veeva Vault CRM now positions itself as a unified life-sciences CRM covering customer profiling, content, territory and account planning, key-account management, sampling, field engagement, call center and inside sales, with offline mobile support and embedded AI.

Salesforce Life Sciences Cloud similarly covers HCP/HCO management, territory and product alignment, cycle plans, pre-call planning, customer recommendations, engagement execution, consent, samples, intelligent content and analytics.

IQVIA OCE emphasizes orchestration between personal and non-personal engagement, AI/ML-driven recommendations, customer timelines, analytics, reporting, sample management and expenditure management.

Exeevo adds AI-assisted workflows, route optimization, offline operation, activity capture, mobile CRM, customer insights, next-best actions and voice-to-text.

Therefore the product architecture proposed in this document is intentionally broader than a traditional SFA application.

---

2. Product Vision

2.1 Vision

Build a configurable pharmaceutical field-force platform that gives every employee exactly the information and workflows required for their role while giving management a reliable, auditable view of:

> **Who is working, where they are working, whom they are engaging, what they are doing, what outcomes are being produced, and what should happen next.**

---

3. Core Product Principles

The system should follow these principles.

P1. Mobile-first

A medical representative spends most of their working day away from a desk.

The mobile application is therefore a primary product, not a secondary client.

P2. Offline-first

The field user must remain productive without internet access.

Veeva and Exeevo both explicitly support offline field operation, and this should be treated as an industry-level requirement rather than a luxury.

P3. Configuration over hardcoding

Business rules should be configurable.

Examples:

- Number of visits per doctor
- Call duration
- Allowed visit types
- Sample restrictions
- Required fields
- Territory hierarchy
- Approval levels
- Expense limits
- Leave approval chains
- GPS accuracy requirements
- Working days
- Call frequencies
- Product availability
- Customer classifications

P4. Auditability

Every meaningful business action should leave an audit trail.

P5. Role-based experience

A representative, area manager, regional manager, HR administrator and finance administrator should not see the same application.

P6. Pharma-specific data model

Do not build a generic CRM and sprinkle “Doctor” labels over it.

The data model should inherently understand:

- HCP
- HCO
- Specialty
- Practice
- Hospital
- Clinic
- Pharmacy
- Distributor
- Stockist
- Territory
- Product
- Brand
- Molecule
- Campaign
- Cycle
- Call
- Sample
- Consent
- Medical inquiry
- Adverse event
- KOL
- MSL
- Scientific engagement

P7. Compliance by design

Compliance should be built into workflows, rather than being something added after the screens are finished.

For example, Veeva's sample workflow can validate product, license and signature conditions before allowing a call with samples to be submitted.

---

4. Competitive Analysis

4.1 Competitive Landscape

The major relevant competitors can be divided into several groups.

Platform	Primary Strength	Key Capabilities
Veeva Vault CRM	Pharma-native enterprise CRM	Territories, call reporting, cycle plans, samples, content, consent, medical, events, offline

Platform	Primary Strength	Key Capabilities
Salesforce Life Sciences Cloud	Enterprise CRM + ecosystem	HCP/HCO, territories, activity plans, samples, consent, content, analytics, AI
IQVIA OCE	Life-sciences customer engagement	Omnichannel orchestration, AI/ML, field execution, analytics, customer insights
Exeevo	Unified life-sciences CRM	CRM, medical, marketing, events, AI, mobile, offline, route planning
Pitcher	Digital sales / engagement	Field sales, content presentation, engagement and digital detailing
Cegedim ecosystem	Life-sciences data + CRM	HCP master data, commercial intelligence and field-force capabilities
BeatRoute	SFA / distribution / pharma execution	Field execution, journey planning, productivity, AI-oriented SFA

The strategic lesson is important:

The product opportunity is not “another attendance app.”

The opportunity is a modular Life Sciences Commercial Operations Platform.

---

5. Veeva Analysis

Veeva is the most important benchmark because its product model reflects pharma-specific processes rather than generic CRM.

Vault CRM covers:

- Customer profiling
- Content management
- Territory planning
- Account planning
- Key account management
- Sampling
- Field engagement
- Medical CRM
- Events
- Consent
- Approved email
- Call reporting
- Offline mobile
- AI

5.1 Territory Model

Territories are hierarchical, and territory managers can assign accounts and users to territories.

Requirement

Our application must support:

```
Company
 └── Country
      └── Region
           └── Area
                └── Territory
                     └── Sub-territory
                          └── Representative
```

The hierarchy must be configurable.

---

6. Veeva Cycle Planning Insight

Veeva's cycle planning concept is particularly important.

A cycle defines:

- Name
- Start date
- End date
- Territory

Cycle targets then define:

- Target accounts
- Planned calls
- Total planned calls
- Product-detail goals
- Priority products

Targets may be informed by segmentation, sales history, field input and marketing initiatives.

Therefore, the product should distinguish:

Strategic plan

"What should I achieve this quarter?"

Cycle plan

"Which HCPs should I engage during this cycle?"

Tour plan

"Which HCPs will I visit on which days?"

Daily plan

"What exactly am I doing today?"

Call report

"What actually happened?"

This hierarchy is fundamental.

---

7. Salesforce Life Sciences Cloud Analysis

Salesforce's current Life Sciences offering covers:

- HCP/HCO accounts
- Customer profiles
- Segmentation
- Affiliations
- Territory alignment
- Product-territory alignment
- Objectives
- Cycle plans
- Pre-call planning
- Customer recommendations
- Activity scheduling
- Remote engagement
- Visit management
- Consent
- Sample management
- Content library
- eDetailing
- Compliant email
- Analytics
- Data partnerships

Salesforce's implementation guidance explicitly calls for:

- HCP/HCO data model
- Hierarchies
- Affiliations
- Referral networks
- Third-party customer data
- Segmentation
- Targeting
- Consent
- Data governance
- Customer master data quality
- Sample integration
- Territory design
- Activity planning
- KPIs

This is valuable because it tells us what a proper requirements-gathering exercise should ask before development even begins.

---

8. IQVIA OCE Analysis

IQVIA's OCE model introduces a more sophisticated idea:

Engagement orchestration

Instead of:

> Sales rep visits doctor → logs visit

the system should understand:

```
HCP
 ↓
Previous interactions
 ↓
Digital interactions
 ↓
Previous detailing
 ↓
Sales / prescribing signals
 ↓
Segment
 ↓
```

```
Current objectives
  ↓
Recommended action
  ↓
Rep interaction
  ↓
Captured response
  ↓
Next recommended action
```

IQVIA describes OCE as connecting sales, marketing, medical and other customer-facing functions and supporting personalization, customer interaction timelines, AI-based next-best actions and common life-sciences data.

An IQVIA case study also emphasizes:

- Optimization of field tactics
- Actionable intelligence
- Automated KPI tracking
- Ease of implementation
- Flexibility
- AI/ML recommendations
- Reporting
- Sample management
- Expenditure management
- Integration

---

9. Exeevo Analysis

Exeevo is especially relevant for mobile architecture.

Its mobile application includes:

- AI assistant
- Voice-to-text
- Workflow suggestions
- Activity aggregation
- Route planning
- Location capabilities
- Check-in/check-out
- Offline operation
- Call plans
- Next-best actions
- Product recommendations

- Forecasting
- Analytics
- Scientific plans for MSLs

Its route planning considers:

- Distance
- Appointments
- Spontaneous visits
- Customer locations

and provides navigation and check-in/check-out.

This should directly influence our mobile architecture.

---

10. Product Scope

The complete system should contain the following major components.

```
                    PHARMA FIELD FORCE PLATFORM
                                │
          ┌─────────────────────┼─────────────────────┐
          │                     │                     │
      MOBILE APP           WEB PORTAL              BACKEND
          │                     │                     │
          ├─ Daily Work         ├─ Admin              ├─ API
          ├─ Tour Plan          ├─ HR                 ├─ Auth
          ├─ Visits             ├─ Sales              ├─ Rules
          ├─ HCPs               ├─ Marketing          ├─ Sync
          ├─ Samples            ├─ Finance             ├─ Notifications
          ├─ Expenses           ├─ Analytics           ├─ Audit
          ├─ Attendance         ├─ Compliance          └─ Integrations
          ├─ Content            └─ Configuration
          └─ Maps
```

---

11. User Roles

The system should support configurable roles.

11.1 Corporate Administrator

Responsibilities:

- Organization setup
- User administration
- Permissions
- Master data
- Configuration
- Security
- Business rules
- Integrations

11.2 HR Administrator

- Employee records
- Employee joining
- Transfers
- Leave
- Attendance policies
- Working schedules
- Holidays
- Organizational assignments

11.3 Sales Representative / Medical Representative

- Daily work
- Tour plan
- HCP visits
- Call reports
- Product detailing
- Samples
- Orders
- Expenses
- Attendance
- Customer updates
- Insights

11.4 Area Sales Manager

- Team management
- Team activity
- Tour-plan approval
- Joint working
- Coaching
- Performance
- Location visibility

- Exceptions
- Approval workflows

11.5 Regional Sales Manager

- Multiple areas
- Performance
- Territory analysis
- Regional targets
- Manager performance

11.6 National Sales Manager

- National performance
- Product performance
- Region comparison
- Strategic planning

11.7 Marketing

- Products
- Campaigns
- Messaging
- Content
- Segmentation
- Targeting
- Cycle plans

11.8 Medical Affairs / MSL

- Scientific plans
- KOL management
- Scientific engagements
- Medical inquiries
- Medical insights
- Adverse-event routing

Veeva's Medical CRM similarly separates business administrators, Medical Affairs users and MSLs, with capabilities for scientific interests, territory strategy, KOL relationships, interactions and scientific insights.

11.9 Finance

- Expense validation
- Expense approval
- Reimbursements
- Budgeting

11.10 Compliance

- Audits
- Content approvals
- Sample compliance
- Consent
- Monitoring
- Exception investigations

---

12. Organogram Module

12.1 Requirements

The system shall support organizational structures containing:

- Company
- Division
- Department
- Business Unit
- Country
- Region
- Area
- Territory
- Team
- Employee

12.2 Employee Assignment

Each employee should support:

- Employee ID
- Name
- Profile photo
- Phone
- Email
- Job title
- Role
- Manager
- Department
- Region
- Area
- Territory
- Joining date
- Employment status

- Working schedule
- Assigned products
- Assigned customers
- User permissions

12.3 Org Tree

Users with permission can view:

```
National Sales Manager
       │
  ┌────┼─────┐
Region A   Region B
   │          │
 Areas       Areas
   │
Managers
   │
Representatives
```

12.4 Organizational Changes

Must support:

- Transfer
- Promotion
- Demotion
- Manager change
- Territory reassignment
- Temporary assignment
- Employee deactivation
- Employee reactivation
- Effective dating

Historical data must remain associated with the organization structure that existed at the time.

---

13. Authentication

FR-AUTH-001

The system shall support username/password authentication.

FR-AUTH-002

Optional authentication methods:

- OTP
- MFA
- Authenticator apps
- SSO
- Microsoft Entra ID
- Google Workspace
- SAML
- OAuth 2.0 / OIDC

FR-AUTH-003

The system shall support:

- Password policy
- Password expiration
- Login attempt throttling
- Account lockout
- Session expiration
- Device registration
- Device revocation

FR-AUTH-004

Administrators shall be able to terminate all sessions for an employee.

---

14. Role-Based Access Control

Permissions must operate at multiple levels.

Module level

Can the user access:

- HCPs?
- Samples?
- Expenses?

Action level

Can they:

- View?
- Create?
- Edit?
- Delete?
- Approve?
- Reject?
- Export?

Record level

Can the user see:

- Their records?
- Their team?
- Their region?
- Entire organization?

Field level

Example:

A representative may see an HCP's profile but not financial performance.

A national manager may see territory-level revenue.

---

15. Customer Master

This is one of the most important systems in the entire platform.

The customer model should distinguish:

HCP

Healthcare Professional.

Examples:

- Doctor
- Pharmacist
- Nurse
- Dentist

- Other healthcare professional

HCO

Healthcare Organization.

Examples:

- Hospital
- Clinic
- Pharmacy
- Healthcare network
- Institution

Commercial Entities

- Distributor
- Stockist
- Wholesaler
- Retail pharmacy

---

16. HCP Data Model

The HCP record should support:

- HCP ID
- External IDs
- Name
- Gender
- Specialty
- Sub-specialty
- Qualifications
- Registration/license number
- License expiry
- Phone
- Email
- Preferred language
- Preferred channel
- Consent status
- Addresses
- Practice locations
- Hospital affiliations
- Clinic affiliations
- Pharmacy affiliations
- KOL status
- Segmentation

- Potential score
- Priority
- Product interests
- Communication preferences
- Visit frequency
- Last visit
- Next planned visit
- Notes
- Insights
- Historical calls

The requirement for affiliations and HCP/HCO hierarchies is explicitly reflected in Salesforce's Life Sciences implementation guidance.

---

17. HCO Data Model

Support:

- HCO ID
- Name
- Type
- Ownership
- Registration details
- Addresses
- Departments
- Doctors
- Pharmacies
- Procurement contacts
- Purchase contacts
- Affiliated HCPs
- Business potential
- Customer segment
- Territory
- Geographic coordinates

---

18. Duplicate Management

The system shall detect possible duplicates.

Example:

```
Dr. Muhammad Ali
Muhammad Ali, MBBS
Dr Muhammad Ali
M. Ali
```

Possible duplicate algorithm should consider:

- Name
- Phone
- Registration number
- Email
- Address
- Specialty
- Practice

Duplicates should enter a merge/review workflow.

---

19. Territory Management

Territories may be based on:

- Geography
- City
- Postal code
- District
- GPS polygon
- HCP assignment
- Product
- Specialty
- Sales team
- Business unit

The system shall support:

```
Country
→ Region
→ Area
→ Territory
→ Customer
```

or alternative structures.

Territories must be effective-dated.

20. Geographic Assignment

The platform should support geospatial territory definitions.

A territory can be:

- Polygon
- Radius
- Administrative boundary
- List of locations

Example:

```
Lahore Central
 ├── Gulberg
 ├── Model Town
 ├── Garden Town
 └── Jail Road
```

21. Customer Segmentation

Segmentation can use:

- A/B/C
- Gold/Silver/Bronze
- Tier 1/2/3
- Potential
- Prescription volume
- Specialty
- Revenue
- Strategic importance
- KOL status

Segmentation must be configurable.

22. Targeting

The system shall allow managers to create customer targets based on:

- Segment

- Product
- Specialty
- Territory
- Sales performance
- Visit frequency
- Prescription potential
- Strategic importance
- Campaign
- New product launch

---

23. Product Master

Products should have:

- Product ID
- Brand name
- Generic/molecule
- Strength
- Form
- Packaging
- SKU
- Manufacturer
- Product category
- Therapeutic class
- Indication
- Regulatory status
- Launch date
- Discontinuation date
- Price
- Sample status
- Stock status
- Promotional status

Hierarchy:

```
Therapeutic Area
     ↓
Molecule
     ↓
Brand
     ↓
Product
```

```
     ↓
SKU / Pack
```

---

24. Product Assignment

Products must be assigned to:

- Reps
- Territories
- Teams
- Regions
- Campaigns
- Cycles

---

25. Cycle Planning

This should be a central module.

Cycle

Example:

```
Cycle:
Q3 Cardiology
Start: 01-Jul
End: 30-Sep
Territory: Lahore Central
```

Targets

```
Dr. A
Required Visits: 8

Dr. B
Required Visits: 6

Dr. C
Required Visits: 4
```

Product goals

```
Product A: 5 details
Product B: 4 details
Product C: 3 details
```

Veeva's cycle model explicitly supports target accounts, planned call counts and product-detail goals.

---

26. Tour Plan

This is one of the most important mobile workflows.

A **Tour Plan** is the operational schedule generated from a cycle plan.

Example:

```
Cycle Plan
    ↓
Target HCPs
    ↓
Customer availability
    ↓
Geography
    ↓
Frequency
    ↓
Working days
    ↓
Rep availability
    ↓
Route optimization
    ↓
Tour Plan
    ↓
Manager approval
    ↓
Daily execution
```

---

27. Tour Plan Requirements

A tour plan should allow:

- Daily planning
- Weekly planning
- Monthly planning
- Multiple visits per day
- HCP selection
- HCO selection
- Unplanned visits
- Joint visits
- Leave
- Training
- Meetings
- Travel
- Holidays
- Office work

---

28. Tour Plan Creation Flow

User Flow: Create Monthly Tour Plan

Step 1

Representative opens:

Planning → Tour Plan → Create

Step 2

Select:

- Month
- Territory
- Working schedule

Step 3

System displays target customers.

Example:

```
Priority A
  Dr. Ahmed       8 visits
  Dr. Sara        8 visits

Priority B
  Dr. Ali         4 visits
  Dr. Khan        4 visits
```

Step 4

System recommends visits based on:

- Required frequency
- Doctor availability
- Territory
- Geographic distance
- Existing appointments
- Working days

Step 5

Representative can manually move visits.

Example:

```
Monday
09:00 Dr. Ahmed
10:30 Dr. Sara
12:00 Dr. Ali

Tuesday
09:00 Dr. Hassan
10:30 Dr. Bilal
```

Step 6

System calculates:

- Distance
- Estimated travel time
- Daily workload
- Overlapping appointments
- Missed target risks

Step 7

Representative submits.

Status:

```
Draft
   ↓
Submitted
   ↓
Manager Review
   ↓
Approved
```

or:

```
Rejected
   ↓
Edit
   ↓
Resubmit
```

---

29. Manager Tour Plan Approval

The manager should see:

- Rep
- Territory
- Planned visits
- Target coverage
- Frequency adherence
- Travel distance
- Working days
- Leaves
- Exceptions
- Unplanned visits

Manager actions:

- Approve
- Reject
- Request changes

- Edit
- Add visit
- Remove visit

Every decision should be audited.

---

30. Daily Work

The mobile home page should effectively answer:

> “What do I need to do today?”

Example:

```
GOOD MORNING

Tuesday, 19 Aug

Attendance
✓ Checked in

Today's target
6 visits

Completed
2 / 6

Next Visit
Dr. Ahmed
10:30 AM

Distance
2.4 km

Pending
1 expense
1 follow-up
```

---

31. Daily Activity Types

Support:

- Customer visit
- Follow-up
- Phone call
- Video call
- Event
- Training
- Office meeting
- Travel
- Leave
- Administrative work
- Joint field work
- Distributor visit
- Pharmacy visit
- Stockist visit

---

32. Attendance

Attendance requirements should include:

- Check in
- Check out
- GPS coordinates
- Timestamp
- Device
- Accuracy
- Network status
- Optional selfie
- Optional biometric verification
- Attendance location
- Late arrival
- Early departure
- Working hours

---

33. GPS Attendance

The organization should define:

- Allowed radius

- Accuracy threshold
- Whether GPS is mandatory
- Whether mock locations are prohibited
- Whether location permission is required during work hours
- Whether background location is permitted

Do not permanently track employees simply because GPS exists.

Location collection must be purpose-driven and configurable.

---

34. GPS Anti-Fraud

Potential detection:

- Mock GPS
- Emulator
- Impossible travel
- Teleportation
- GPS spoofing
- Multiple account/device usage
- Excessive accuracy anomalies
- Repeated check-in from identical coordinates
- Device clock manipulation

These should preferably create **exceptions**, rather than automatically accusing users.

---

35. Visit / Call Management

The call object should contain:

```
Call
 ├── User
 ├── Account
 ├── HCP
 ├── HCO
 ├── Territory
 ├── Date
 ├── Start time
 ├── End time
 ├── Location
 ├── Call type
 ├── Objective
```

```
├── Products
├── Content
├── Samples
├── Orders
├── Discussion
├── Response
├── Insights
├── Follow-up
└── Attachments
```

Veeva's call model similarly separates a parent call record from call details, attendees and product-detail records.

---

36. Visit Types

Configurable:

- Planned
- Unplanned
- Follow-up
- New customer
- Existing customer
- Joint call
- Group call
- Remote
- Phone
- Video
- Event
- Medical
- Commercial

---

37. Pre-Call Planning

Before visiting a doctor, the application should show:

- Customer profile
- Previous call
- Previous discussion
- Previous products
- Samples previously provided
- Open follow-ups
- Latest insights

- Customer preferences
- Target products
- Recommended objective
- Outstanding commitments
- Previous objection
- Suggested next action

Veeva explicitly supports pre-call notes, next-call notes and key messages, reinforcing this as an appropriate call workflow.

---

38. Check-In Workflow

```
Open today's visit
        ↓
Navigate
        ↓
Arrive
        ↓
Check In
        ↓
GPS validation
        ↓
Pre-call screen
        ↓
Customer interaction
        ↓
Detail products/content
        ↓
Samples
        ↓
Capture response
        ↓
Create follow-up
        ↓
Check Out
        ↓
Submit Call Report
```

---

39. Geo-Validation

On check-in:

```
Current GPS
      ↓
Customer location
      ↓
Calculate distance
      ↓
Within allowed radius?
```

Possible outcomes:

Valid

Allow normal workflow.

Borderline

Show warning.

Invalid

Allow configurable outcomes:

- Block check-in
- Require explanation
- Require manager approval
- Mark as exception
- Allow with audit flag

This must never be globally hardcoded.

---

40. Call Report

Required sections:

Header

- HCP
- HCO
- Location
- Territory
- Call type
- Date/time

Objective

- Planned objective
- Actual objective

Products Discussed

- Product
- Message
- Response
- Interest level

Content

- Presentation
- Brochure
- Video
- Visual aid
- Document

Samples

- Product
- Quantity
- Batch
- Expiry
- Signature/consent where required

Insights

- Customer response
- Competitor information
- Market information
- Product feedback
- Clinical insight

Follow-up

- Task
- Owner
- Due date

---

41. Call Submission

Submission should run business rules before saving as final.

Example:

```
Has samples?
    ↓
Check sample rules
    ↓
Valid license?
    ↓
Valid consent?
    ↓
Signature required?
    ↓
Required fields present?
    ↓
GPS valid?
    ↓
Submit
```

Veeva uses validation rules to ensure sample, license and signature requirements are satisfied before a call containing samples can be submitted.

After submission:

- Call becomes read-only
- Audit record generated
- Transactions generated
- KPIs updated
- Sync event generated

---

42. Sample Management

This should be a dedicated subsystem, not just a numeric field inside the call report.

Inventory

Track:

- Product
- Batch
- Expiry
- Quantity
- Warehouse
- Rep
- Territory
- Stock transfer

- Distribution
- Adjustment
- Return
- Destruction

Sample transaction

```
Warehouse
  ↓
Rep inventory
  ↓
HCP
  ↓
Call
  ↓
Sample transaction
```

---

43. Sample Compliance

Depending on jurisdiction/product, support:

- HCP eligibility
- License validation
- Sample opt-in
- Consent
- Signature
- Paper consent reference
- Controlled substance restrictions
- Quantity limits
- Frequency restrictions
- Expiry checks
- Batch tracking

Veeva's sampling implementation demonstrates that consent can have expiration dates and that controlled substances can require additional signatures even when a general sample opt-in remains valid.

---

44. Electronic Signature

Support:

- On-device signature
- Remote signature

- QR code
- Share link
- Signature timestamp
- User/device metadata
- Consent text version
- Document version
- Signature evidence

Veeva supports remote sample opt-in through share links and QR codes, illustrating the value of separating signature capture from the representative's physical device.

---

45. Consent Management

Consent should be granular.

Example:

```
HCP
 ├── Email
 │     ├── Product A
 │     └── Product B
 ├── SMS
 ├── WhatsApp
 ├── Phone
 ├── CLM Tracking
 ├── Sampling
 └── Events
```

Track:

- Opt-in
- Opt-out
- Date
- Source
- Channel
- Purpose
- Language
- Consent version
- Expiry
- Capture user
- Evidence

Veeva's Consent Capture explicitly supports channel-specific and purpose-specific consent rather than treating consent as one global flag.

46. Content Management

The application should support approved content.

Content types:

- PDF
- Video
- Presentation
- Interactive detail aid
- Image
- Brochure
- Product information
- Scientific document
- FAQ
- Email template

Content must have:

- Version
- Status
- Effective date
- Expiry date
- Country
- Language
- Product
- Specialty
- Audience
- Approval status

---

47. Content Workflow

```
Draft
  ↓
Medical Review
  ↓
Regulatory Review
  ↓
Compliance Approval
  ↓
Published
  ↓
```

```
Mobile Sync
 ↓
Rep Uses Content
 ↓
Usage Analytics
 ↓
Retire
```

Veeva's CLM architecture uses centrally managed approved content and tracks content use during interactions.

---

48. Content Analytics

Track:

- Content opened
- Slides viewed
- Time spent
- Completion rate
- Most-used content
- Content by specialty
- Content by product
- Content effectiveness
- Last version used

---

49. Approved Communications

Potential channels:

- Email
- SMS
- WhatsApp integration where legally/technically appropriate
- Push
- In-app notifications

Communications must respect consent.

Veeva's Approved Email system performs recipient and consent filtering before sending.

---

50. Events Management

Support:

- Event creation
- Speaker management
- Venue
- Budget
- Attendees
- Registration
- Invitations
- Approval
- QR registration
- Attendance
- Expenses
- Feedback
- Follow-up

Veeva supports event rules, approvals, self-registration, attendee management and automated communications.

---

51. Medical Affairs

Do not force Medical Affairs into the exact same workflow as sales.

Medical users should have:

- Scientific plans
- KOL management
- Scientific interests
- Medical interactions
- Medical insights
- Medical inquiries
- Case management
- Follow-up tasks

Veeva explicitly provides Medical CRM features for scientific engagement and medical inquiries.

---

52. Adverse Event / Pharmacovigilance

This is a critical pharma requirement.

A rep may hear:

> “Doctor says patient experienced rash after taking Product X.”

The app must allow the representative to classify this as a potential safety event and route it into a controlled workflow.

The system should not require the representative to determine medical causality.

---

53. Adverse Event Flow

```
Rep hears safety information
        ↓
Select "Report Safety Information"
        ↓
Minimal required information
        ↓
Patient information where appropriate
        ↓
Product information
        ↓
Event description
        ↓
Reporter information
        ↓
Attachments
        ↓
Immediate submission
        ↓
Pharmacovigilance queue
        ↓
Safety team review
        ↓
External reporting / case processing
```

For Pakistan specifically, DRAP maintains pharmacovigilance guidelines and provides adverse-event reporting channels. DRAP states that industry product registration holders can submit electronic adverse-event reports using E2B XML, with CIOMS as an alternative route in specified circumstances.

Therefore the platform should architect a **PV integration boundary**, rather than embedding an unchangeable country-specific process into the core mobile application.

---

54. Medical Inquiry

A field user should be able to create:

```
Medical Inquiry
 ├── HCP
 ├── Question
 ├── Product
 ├── Category
 ├── Urgency
 ├── Attachment
 └── Assigned Medical Team
```

Workflow:

```
Created
  ↓
Assigned
  ↓
In Review
  ↓
Response Prepared
  ↓
Medical Approval
  ↓
Responded
  ↓
Closed
```

---

55. Order Management

Depending on business model, support:

- Product
- SKU
- Quantity
- Price
- Discount
- Distributor
- Pharmacy
- Delivery address

- Order status
- Approval
- ERP synchronization

Statuses:

```
Draft
Submitted
Approved
Rejected
Sent to ERP
Confirmed
Dispatched
Delivered
Cancelled
```

---

56. Distributor / Stockist Management

Include:

- Distributor master
- Stockist master
- Credit limit
- Territory
- Products
- Stock
- Sales
- Orders
- Returns
- Payment status
- Contracts
- Contact persons

This is especially useful where pharmaceutical sales involve distributor/wholesaler channels.

---

57. Retail Pharmacy

Support:

- Pharmacy master
- Location
- Owner

- Pharmacist
- Contact
- Product availability
- Stock
- Orders
- Competitor information
- Visits

---

58. Competitor Intelligence

Field representatives should be able to capture:

- Competitor product
- Price
- Availability
- Promotion
- Market feedback
- Doctor preference
- Competitor claims
- Competitor samples
- Promotional material

These observations should flow into dashboards.

---

59. Field Insights

Create a structured insight model.

Examples:

```
Product Insight
Competitor Insight
Market Insight
Customer Insight
Supply Insight
Pricing Insight
Clinical Insight
Regulatory Insight
```

Fields:

- Category

- Description
- Customer
- Product
- Territory
- Date
- Severity
- Confidence
- Attachment
- Follow-up owner

---

60. Expense Management

Represent expenses as:

```
Expense
  ├── Date
  ├── Category
  ├── Amount
  ├── Currency
  ├── Location
  ├── Customer/Event
  ├── Receipt
  ├── Description
  ├── Policy status
  └── Approval status
```

Categories:

- Fuel
- Mileage
- Hotel
- Food
- Toll
- Parking
- Transportation
- Customer event
- Meeting
- Other

---

61. Expense Approval

```
Submitted
 ↓
Policy engine
 ↓
Within policy?
 ├── Yes → Manager
 └── No  → Exception
                ↓
             Finance review
```

---

62. Leave Management

Support:

- Leave types
- Leave balances
- Leave requests
- Supporting documents
- Approval hierarchy
- Calendar
- Half-day
- Full-day
- Cancellation
- Holiday calendar

Leave must interact with tour planning.

A representative on approved leave should not accidentally retain six planned customer visits that day.

---

63. Joint Working

Manager and rep can perform a joint visit.

Example:

```
Rep
 + Area Manager
      ↓
```

```
Visit HCP
      ↓
Rep records activity
      ↓
Manager adds coaching
      ↓
Post-call feedback
```

Support:

- Joint-work request
- Approval
- Shared itinerary
- Participant records
- Coaching notes
- Evaluation form

---

64. Coaching

Managers should have a structured coaching workflow.

Example:

```
Pre-call
  ↓
Observe visit
  ↓
Evaluate
  ↓
Strengths
  ↓
Improvement areas
  ↓
Action plan
  ↓
Follow-up date
```

Scoring categories:

- Preparation
- Opening
- Need discovery
- Product knowledge

- Message delivery
- Objection handling
- Closing
- Compliance
- Data capture

---

65. Performance Management

KPIs should include:

Activity

- Calls planned
- Calls completed
- Call adherence
- Productive calls
- Average calls/day

Coverage

- HCP coverage
- Target coverage
- Frequency adherence
- Segment coverage

Quality

- Call duration
- Detail quality
- Product discussions
- Follow-ups
- Insights

Commercial

- Orders
- Sales
- Growth
- Target achievement

Operational

- Attendance
- Travel
- Route efficiency
- Expenses

66. Do Not Build a "Calls per Day" Culture

The dashboard should avoid rewarding meaningless activity.

A representative making 12 low-quality visits should not automatically outrank someone making 7 high-value interactions.

The platform should progressively support outcome-linked metrics.

This aligns with the current SFA trend toward outcome-linked field execution rather than purely activity-based effectiveness measurement.

67. Dashboards

Rep Dashboard

- Today's visits
- Completion
- Monthly target
- Coverage
- Pending tasks
- Expenses
- Follow-ups

Area Manager

- Team attendance
- Team calls
- Tour adherence
- Coverage
- Rep ranking
- Exceptions
- GPS anomalies
- Expense anomalies

Regional Manager

- Area comparison
- Product performance
- Target achievement
- Customer coverage
- Trend

National Manager

- National performance
- Region performance
- Product performance
- Territory opportunity
- Forecast
- Sales correlation

---

68. Map Dashboard

Map layers:

- Team members
- HCPs
- HCOs
- Pharmacies
- Visits
- Planned route
- Completed visits
- Missed calls
- Exception locations

---

69. Notifications

Support:

- Push
- In-app
- Email
- SMS where required

Triggers:

- Tour approval
- Tour rejection
- New assignment
- Visit reminder
- Missed visit
- Expense rejection
- Leave approval
- Sample shortage
- Content update

- Medical inquiry
- Follow-up
- Manager message

---

70. Task Management

Any process should be able to generate a task.

Example:

```
Call
 ↓
Follow-up required
 ↓
Task created

Task:
Call Dr. Ahmed
Due: 22 Aug
Owner: Rep
Priority: High
```

Task types:

- Customer follow-up
- Manager action
- Medical inquiry
- Expense correction
- Sample reconciliation
- Event task

---

71. Calendar

Calendar must display:

- Visits
- Tour plan
- Leave
- Training
- Meetings
- Events
- Travel

- Tasks

Views:

- Day
- Week
- Month

---

72. Route Optimization

Input:

```
Customer locations
+
Appointment windows
+
Working hours
+
Travel mode
+
Traffic
+
Priority
+
Required frequency
```

Output:

```
Optimized itinerary
```

The route engine should support manual override.

Never force the “optimal” route over human knowledge.

---

73. Spontaneous Visit

Rep should be able to:

```
Open HCP
  ↓
```

```
Record Visit
 ↓
System checks:
  planned?
  territory?
  duplicate?
  GPS?
 ↓
Create unplanned call
```

Require configurable reason:

- Customer requested
- Opportunity
- Follow-up
- Nearby
- Manager instruction
- Other

---

74. Offline Architecture

This is a major technical requirement.

The mobile application needs:

```
                 MOBILE
                    │
         ┌──────────┴─────────┐
         │                    │
    Local DB          Sync Engine
         │                    │
         └──────────┬─────────┘
                    │
                API Layer
                    │
                 Backend
                    │
               PostgreSQL
```

---

75. Offline Data

Cache selectively:

Always available

- User
- Role
- Assigned HCPs
- Assigned HCOs
- Products
- Current tour plan
- Recent calls
- Current content
- Tasks
- Reference data

Optional

- Historical calls
- Historical sales
- Large media files

---

76. Sync Engine

The application must support:

- Initial synchronization
- Incremental synchronization
- Background synchronization
- Manual sync
- Retry
- Conflict handling
- Partial sync
- Failed record queue
- Upload queue
- Download queue
- Sync status

---

77. Sync States

Every local record can have:

```
LOCAL
PENDING_UPLOAD
SYNCING
SYNCED
CONFLICT
FAILED
```

---

78. Conflict Resolution

Possible conflict:

Rep edits HCP phone offline.

Manager changes same phone online.

System should not silently overwrite.

Use:

```
Conflict detected
  ↓
Original value
Server value
Local value
  ↓
Resolution rule / authorized user
```

---

79. Offline Security

Sensitive data cached on device must be:

- Encrypted
- Access-controlled
- Expirable
- Revocable
- Deleted on logout/device revocation where appropriate

OWASP MASVS specifically covers mobile storage, cryptography, authentication, network communication, platform interaction, code security, resilience and privacy.

---

80. Backend Architecture

Recommended initial architecture:

```
                   API Gateway
                        │
         ┌──────────────┼──────────────┐
         │              │              │
   Auth Service   Core API       Notification
         │              │              │
         │        Business Services    │
         │              │              │
         └──────────────┼──────────────┘
                        │
                  PostgreSQL
                        │
           ┌────────────┼────────────┐
           │            │            │
         Redis        Object       Search
                      Storage      Engine
```

---

81. Recommended Backend Modules

Start modular.

Identity Service

Authentication, sessions, roles.

Organization Service

Employees, hierarchy, territory.

Customer Service

HCP/HCO master.

Product Service

Brands and SKUs.

Planning Service

Cycles, tour plans, schedules.

Field Activity Service

Calls, visits, tasks.

Sample Service

Inventory and transactions.

Expense Service

Claims and reimbursement.

Content Service

Approved documents.

Consent Service

Consent and preferences.

Notification Service

Push/email/SMS.

Analytics Service

Metrics and reporting.

Audit Service

Immutable business audit history.

---

82. Database Model

Core tables/entities:

```
users
roles
permissions
employees
```

```
organizations
departments
regions
areas
territories
territory_assignments

hcp
hco
hcp_hco_affiliations
hcp_addresses
hcp_specialties
hcp_segments

products
product_brands
product_skus
product_assignments

cycles
cycle_targets
cycle_products

tour_plans
tour_plan_days
tour_plan_items

activities
calls
call_products
call_attendees
call_content
call_samples
call_insights

sample_inventory
sample_transactions
sample_batches

orders
order_lines

expenses
expense_lines
expense_attachments

leave_requests
attendance_records
```

```
location_events

tasks
notifications

content
content_versions
content_approvals

consents
consent_events

medical_inquiries
adverse_events

events
event_attendees
event_expenses

audit_logs
sync_records
integration_logs
```

---

83. API Requirements

The backend should expose REST or GraphQL APIs.

Core endpoints:

```
POST /auth/login
POST /auth/refresh
POST /auth/logout

GET /me
GET /me/permissions

GET /hcp
GET /hcp/{id}
POST /hcp
PUT /hcp/{id}

GET /hco
GET /hco/{id}
```

```
GET /products
GET /territories

GET /cycles
GET /cycles/{id}

GET /tour-plans
POST /tour-plans
PUT /tour-plans/{id}
POST /tour-plans/{id}/submit
POST /tour-plans/{id}/approve
POST /tour-plans/{id}/reject

GET /calls
POST /calls
PUT /calls/{id}
POST /calls/{id}/submit

POST /attendance/check-in
POST /attendance/check-out

GET /samples/inventory
POST /samples/transactions

GET /expenses
POST /expenses
POST /expenses/{id}/submit

POST /medical-inquiries
POST /adverse-events

GET /tasks
POST /tasks

POST /sync
GET /sync/status
```

---

84. Idempotency

Critical APIs must be idempotent.

Especially:

- Call submission

- Sample transaction
- Order creation
- Expense submission
- Attendance
- Payment
- Sync

This prevents duplicate records when a mobile device retries after a network failure.

---

85. Audit Trail

Audit record:

```
Audit ID
User
Role
Timestamp
Action
Object
Object ID
Previous value
New value
IP
Device
Location if applicable
Reason
```

Actions:

- Create
- Update
- Delete
- Submit
- Approve
- Reject
- Export
- Login
- Logout
- Permission change
- Data download

---

86. Immutable Records

Certain records should become immutable after final submission.

Examples:

- Submitted call report
- Sample transaction
- Electronic signature
- Approved expense
- Audit record
- Approved content

Corrections should create a correction/amendment trail rather than erase history.

---

87. Electronic Records

If the system is intended for regulated environments, it should be designed to accommodate applicable electronic-record/electronic-signature requirements.

For example, FDA 21 CFR Part 11 covers electronic records created, modified, maintained, archived, retrieved or transmitted under applicable FDA record requirements.

This does not mean that every deployment automatically falls under Part 11. Applicability must be determined according to the customer's regulatory environment and use case.

---

88. Security Requirements

Minimum security baseline:

- TLS 1.2+
- Modern cipher suites
- Secure password hashing
- MFA
- RBAC
- Principle of least privilege
- Token expiration
- Device binding
- Secure mobile storage
- API authorization
- Input validation
- Rate limiting

- WAF
- Security headers
- Encryption at rest
- Secrets management
- Audit logging
- Backup encryption
- Vulnerability scanning

---

89. Mobile Security

Implement against OWASP MASVS.

Areas:

- Secure storage
- Cryptography
- Authentication
- Network security
- Platform interaction
- Code security
- Resilience
- Privacy

For backend APIs, OWASP ASVS should be used as the corresponding web/API security baseline.

---

90. Encryption

In transit

TLS.

At rest

Database encryption.

Mobile

Encrypted local database / secure storage.

Secrets

Never hardcode:

- API keys
- Encryption secrets
- Service credentials
- Signing keys

Use a secrets-management system.

---

91. Privacy

The application may contain:

- Employee location
- Employee performance
- HCP personal information
- Medical information
- Consent
- Signatures
- Regulatory information

Therefore privacy requirements need to be designed before development.

The platform should support:

- Data minimization
- Purpose limitation
- Retention periods
- Access rights
- Deletion/anonymization workflows where legally applicable
- Consent records
- Data export
- Data processing logs

For deployments involving personal/health information in jurisdictions such as the EU, applicable GDPR requirements need to be addressed, especially where health-related data receives heightened protection.

---

92. Pakistan Regulatory Considerations

For Pakistan, the requirements baseline should explicitly include DRAP processes.

DRAP currently lists:

- Ethical Marketing to Healthcare Professionals Rules, 2021
- Pharmacovigilance Rules, 2022
- Advertising guidelines
- Pharmacovigilance guidelines
- Guidance on healthcare-professional adverse-event reporting

DRAP also maintains guidelines around expenditure details related to marketing activities directed toward healthcare professionals.

Therefore a Pakistan deployment should have configurable:

- Event expenditure
- HCP interaction expense
- Promotional activity records
- Sample/distribution records
- Approval workflows
- Audit logs
- Regulatory reports

---

93. Non-Functional Requirements

NFR-001 Availability

Target:

99.9% monthly availability for production backend, excluding planned maintenance.

For larger enterprise contracts:

99.95%+ should be considered.

---

94. NFR-002 Mobile Responsiveness

Standard interactions:

Target:

< 300 ms perceived interaction for locally available data.

API operations:

Target:

< 1 second for common requests under normal conditions.

---

95. NFR-003 Scalability

Architecture should support:

Initial:

- 1 organization
- 1,000 users
- 100,000 HCPs

Target architecture:

- 100,000+ users
- Millions of HCP records
- Millions of activities
- Large content libraries

Do not prematurely microservice every feature.

Use modular architecture first, then separate high-volume services where justified.

---

96. NFR-004 Offline Availability

Core workflows must continue without network:

- Login using previously authenticated session where policy permits
- View assigned customers
- View today's schedule
- View products
- Record visit
- Record call
- Capture notes
- Capture signature
- Create task
- Queue samples
- Queue expenses
- View content already downloaded

---

97. NFR-005 Sync Reliability

The synchronization layer should survive:

- Airplane mode
- Weak signal
- App suspension
- Battery loss
- Application crash
- Duplicate requests
- Server timeout

No user activity should silently disappear.

---

98. NFR-006 Observability

Backend needs:

- Centralized logs
- Metrics
- Tracing
- Error tracking
- API performance monitoring
- Database monitoring
- Queue monitoring
- Sync monitoring
- Alerting

---

99. NFR-007 Backup

Define:

- Daily backups
- Point-in-time recovery
- Backup encryption
- Offsite backup
- Restoration testing

Target:

RPO: ≤ 15 minutes

RTO: ≤ 1 hour

for enterprise-grade deployment.

---

100. Disaster Recovery

Support:

- Database failover
- Backup restoration
- Infrastructure redeployment
- DNS failover where justified
- Disaster recovery runbook
- Regular DR testing

---

101. Localization

Application must support:

- English
- Urdu
- Arabic and additional languages later

Architecture must support:

- RTL
- Translation keys
- Local date formats
- Currency
- Number formats
- Time zones

Do not embed translated strings directly in business logic.

---

102. Multi-Tenant Architecture

If this is intended as a SaaS product:

```
Platform
 ├── Tenant A
```

```
├── Tenant B
├── Tenant C
└── Tenant D
```

Every data record needs a tenant boundary.

Tenant isolation must be enforced server-side.

Never rely solely on frontend filtering.

---

103. Tenant Configuration

Each tenant should configure:

- Organization
- Branding
- Roles
- Modules
- Working days
- Working hours
- Holidays
- Territories
- Call forms
- Sample policies
- Expense rules
- Approval chains
- Languages
- Notification templates
- GPS policies
- Retention

---

104. Business Rules Engine

Do not hardcode business rules inside dozens of Flutter/React screens.

Create configurable rules.

Example:

```
RULE:
IF
```

```
  visit.type = "Sample"
AND
  hcp.sample_license_valid = false
THEN
  block_submission
```

Another:

```
IF
  expense.amount > 10000
THEN
  approval_level = "Regional Manager"
```

Another:

```
IF
  check_in.distance > allowed_radius
THEN
  require_exception_reason
```

---

105. Approval Engine

Generic approval framework:

```
Record
 ↓
Rule evaluation
 ↓
Approval workflow
 ↓
Approver
 ↓
Approve / Reject / Return
 ↓
Next stage
```

Reusable for:

- Tour plans
- Expenses
- Leave

- HCP creation
- HCP correction
- Events
- Discounts
- Samples
- Orders
- Content

---

106. Search

Search must be extremely fast.

Search HCP by:

- Name
- Phone
- Registration
- Specialty
- City
- Hospital
- Territory

Search should tolerate spelling variation.

---

107. Import / Export

Support:

Import

- CSV
- Excel
- API
- SFTP
- JSON
- XML

Export

- Excel
- CSV
- PDF
- API

Imports require:

- Validation
- Preview
- Error reporting
- Duplicate detection
- Rollback
- Import logs

---

108. Master Data Management

Master-data governance must exist for:

- HCP
- HCO
- Product
- Territory
- Employee
- Geography
- Specialty
- Diagnosis/indication
- Distributor
- Pharmacy

Do not allow every mobile user to arbitrarily modify enterprise master data.

Use:

```
Rep Suggests Change
       ↓
Master Data Review
       ↓
Approved
       ↓
Master Record Updated
```

---

109. Integrations

Plan for:

ERP

SAP Oracle Microsoft Dynamics

HR

Employee master Attendance Leave

MDM

HCP data providers Customer data vendors

BI

Power BI Tableau Looker

Communication

Email SMS Push WhatsApp Business where applicable

Maps

Google Maps Apple Maps Mapbox

Identity

Microsoft Entra ID Google SAML/OIDC

Cloud

AWS Azure GCP

---

110. Integration Architecture

Use integration adapters.

```
Core Platform
       │
Integration Layer
       │
 ┌─────┼───────┬───────┐
ERP   HR      MDM     BI
```

Do not tightly couple the business logic to one ERP.

111. Web Administration Portal

The web portal should contain:

Dashboard

- Organization overview
- Activity
- Performance
- Alerts

Organization

- Employees
- Organogram
- Roles
- Permissions
- Territories

Customers

- HCPs
- HCOs
- Affiliations
- Segments

Planning

- Cycles
- Tour plans
- Territories
- Targets

Products

- Brands
- SKUs
- Samples
- Inventory

Content

- Documents
- Approval
- Version control

Operations

- Attendance
- Expenses
- Visits
- Orders

Compliance

- Consent
- Audit logs
- Samples
- Event expenses

Analytics

- Reports
- Dashboards
- Exports

---

112. Mobile Application Navigation

Recommended structure:

```
Home
 │
 ├── Today's Work
 │
 ├── Customers
 │
 ├── Calendar
 │
 ├── Tour Plan
 │
 ├── Visits
 │
 ├── Samples
 │
 ├── Orders
 │
 ├── Expenses
 │
 ├── Tasks
 │
```

```
├── Content
│
└── More
      ├── Attendance
      ├── Medical Inquiry
      ├── Adverse Event
      ├── Events
      ├── Reports
      └── Profile
```

---

113. Mobile Home Screen

The home screen should prioritize actions rather than reports.

Recommended:

```
Good morning, Ahmed

TODAY
6 planned visits
2 completed

NEXT
Dr. Hassan
10:30 AM
1.8 km

[Start Navigation]

TARGET PROGRESS
██████░░ 60%

ATTENDANCE
Checked in 08:52

PENDING
3 Tasks
1 Expense
```

---

114. Representative Daily User Flow

```
Login
 ↓
Home
 ↓
Check attendance
 ↓
Review today's plan
 ↓
Open first customer
 ↓
Review pre-call briefing
 ↓
Navigate
 ↓
Check-in
 ↓
Visit
 ↓
Detail product/content
 ↓
Capture feedback
 ↓
Distribute sample
 ↓
Capture signature if required
 ↓
Create follow-up
 ↓
Check-out
 ↓
Submit call
 ↓
Next visit
```

---

115. Manager Daily Flow

```
Login
 ↓
Dashboard
 ↓
```

```
Team Attendance
 ↓
Exceptions
 ↓
Tour Plan Approvals
 ↓
Team Activities
 ↓
Rep Performance
 ↓
Select Rep
 ↓
Review visits
 ↓
Review missed targets
 ↓
Coach / assign tasks
 ↓
End-of-day review
```

---

116. HCP Creation Flow

```
Rep
 ↓
Add Customer
 ↓
Select HCP
 ↓
Enter:
Name
Specialty
Phone
Clinic
Address
Registration
 ↓
Duplicate Check
 ↓
Possible duplicate?
 ├── Yes → Review
 └── No  → Submit
                ↓
              Approval
```

```
             ↓
        Master record
```

---

117. Tour Plan Detailed Example

Assume:

```
Rep:
Ahmed

Territory:
Lahore Central

Cycle:
Cardio Q3

Required:
Dr A = 8
Dr B = 6
Dr C = 4
```

System calculates:

```
Available working days: 23

Dr A
Week 1: 2
Week 2: 2
Week 3: 2
Week 4: 2

Dr B
Week 1: 2
Week 2: 1
Week 3: 2
Week 4: 1
```

System then considers geography:

```
Gulberg cluster
 ↓
```

```
Dr A
Dr B
Dr C
```

So it groups those visits on the same day.

---

118. Tour Plan Optimization Algorithm

Inputs:

```
Customer priority
Customer frequency
Customer availability
Territory
Distance
Travel time
Appointment windows
Rep working hours
Leaves
Holidays
Existing meetings
```

Objective function could optimize:

```
maximize customer coverage
+
maximize target adherence
+
maximize customer priority
-
travel time
-
distance
-
schedule conflicts
```

The representative must retain manual control.

---

119. AI Roadmap

AI should not be the first feature.

Build the data foundation first.

Stage 1

Rules:

- Missed target alerts
- Route optimization
- Visit reminders
- Sample anomalies
- Expense anomalies

Stage 2

Predictive:

- Best customer to visit
- Best day/time
- Attrition/churn risk
- Coverage gaps
- Sales forecasting

Stage 3

Generative:

- Pre-call briefing
- Call summaries
- Follow-up suggestions
- Voice-to-call-report
- Insight extraction

Stage 4

Agentic:

```
"Plan my next week."
```

AI:

```
Reviews targets
Reviews calendars
Reviews HCP availability
Reviews previous engagement
Optimizes route
Creates draft plan
Explains decisions
Requests user approval
```

The AI should propose actions, not silently alter regulatory/business records.

---

120. Voice-to-Call Report

A future representative could say:

> "Doctor was interested in Product A but requested pricing information. Product B was not discussed. He asked me to follow up next Tuesday."

AI converts this into structured fields:

```
Product A
Interest: High

Product B
Discussed: No

Follow-up:
Pricing information
Due: Tuesday
```

Exeevo already uses voice-to-text and AI-assisted summarization in its mobile life-sciences application.

---

121. Next Best Action

Example:

```
HCP:
Dr. Ali

Previous:
```

```
Product A discussed 3 times

Response:
Positive

No recent Product B discussion

Recommendation:
Discuss Product B

Reason:
Target product
+
High specialty affinity
+
No recent discussion
```

The recommendation should display its reasoning.

---

122. Analytics Event Model

Every meaningful action should generate structured analytics.

Example:

```
event:
call_submitted

properties:
user_id
hcp_id
territory_id
product_ids
planned=true
gps_valid=true
duration
samples_count
content_ids
timestamp
```

This enables future analytics and AI.

---

123. Data Warehouse

For enterprise deployment, operational database and analytics warehouse should eventually be separated.

```
Operational DB
      ↓
CDC / Event Stream
      ↓
Data Warehouse
      ↓
BI / AI
```

This prevents heavy dashboards from destroying transactional performance.

---

124. Core KPI Catalogue

Workforce

- Attendance rate
- Active reps
- Working days
- Leave
- Field time

Customer

- Total HCPs
- Active HCPs
- Target HCPs
- Coverage

Execution

- Planned calls
- Completed calls
- Unplanned calls
- Call adherence
- Frequency adherence

Quality

- Avg call duration
- Product discussion
- Follow-ups

• Insights

Commercial

• Orders
• Sales
• Growth
• Target achievement

Compliance

• Sample exceptions
• Consent failures
• GPS exceptions
• Approval exceptions
• Expired content usage

---

125. Audit and Compliance Dashboard

Compliance administrators should see:

• Invalid sample attempts
• Expired licenses
• Missing signatures
• Out-of-radius visits
• Unapproved content
• Consent violations
• Modified submitted calls
• High expense exceptions
• Suspicious activity
• Inactive users with data-access anomalies

---

126. Fraud / Anomaly Detection

Potential anomalies:

GPS

Rep appears in Karachi at 9:00 and Lahore at 9:20.

Attendance

Same device used by several accounts.

Calls

40 calls entered within 10 minutes.

Samples

Abnormal sample distribution.

Expenses

Repeated identical receipt images.

HCP

Hundreds of HCP records created by a single user.

The system should flag for investigation rather than automatically punish.

---

127. Requirements Traceability

Every requirement should have:

```
Requirement ID
Module
Requirement
Priority
Source
Stakeholder
Acceptance Criteria
Dependencies
Test Case ID
Release
Status
```

Example:

ID	Requirement	Priority	Release
FR-CALL-001	Rep can create call	Must	MVP
FR-CALL-002	Rep can submit call offline	Must	MVP
FR-SAMPLE-001	Record sample transaction	Must	MVP
FR-AI-001	Next-best-action recommendation	Later	R3

128. MoSCoW Prioritization

Must Have

- Authentication
- Users
- Roles
- Organogram
- Territories
- HCP/HCO master
- Products
- Tour plans
- Daily work
- Attendance
- GPS
- Visits
- Call reporting
- Offline storage
- Sync
- Approvals
- Notifications
- Audit logs
- Admin portal
- Basic dashboards

Should Have

- Samples
- Expenses
- Orders
- Content
- Consent
- Events
- Coaching
- Competitor intelligence
- Route optimization

Could Have

- AI summaries
- Voice reports
- Next-best action
- Advanced forecasting
- AI tour-plan optimization

Later

- Fully agentic planning
- Advanced omnichannel orchestration
- Predictive prescriber modeling
- Advanced commercial AI

---

129. MVP

A sensible MVP should include:

Backend

- Authentication
- Organization
- Employees
- Roles
- Territories
- HCP/HCO
- Products
- Tour planning
- Calls
- Attendance
- GPS
- Tasks
- Notifications
- Audit logs
- Offline sync

Mobile

- Login
- Home
- Customers
- Calendar
- Tour Plan
- Daily Work
- Visit
- Call Report
- Attendance
- GPS
- Tasks
- Offline mode

Web

- Dashboard
- Organization
- Users
- Territories
- HCP
- Products
- Tour approval
- Calls
- Reports

---

130. Release 2

Add:

- Samples
- Inventory
- Expenses
- Orders
- Content
- Consent
- Events
- Coaching
- Advanced dashboards
- Route optimization
- HCP segmentation

---

131. Release 3

Add:

- Medical CRM
- Adverse event workflow
- Medical inquiries
- Omnichannel
- AI summaries
- Voice capture
- Next-best action
- Forecasting

---

132. Release 4

Enterprise intelligence:

- AI agentic tour planning
- Predictive customer prioritization
- Sales prediction
- Territory optimization
- Workforce optimization
- Intelligent anomaly detection
- Advanced commercial insights

---

133. What NOT to Build First

Avoid starting with:

- AI chatbot
- Fancy animations
- AI-generated dashboards
- Complicated microservices
- Fully automated route planning
- Advanced predictive models
- Massive event management

Instead build:

```
Identity
  ↓
Organization
  ↓
Customer Master
  ↓
Territory
  ↓
Product
  ↓
Cycle
  ↓
Tour Plan
  ↓
Daily Work
  ↓
Visit
```

```
  ↓
Call
  ↓
Audit
  ↓
Analytics
```

That is the spine of the product.

---

134. Recommended Technical Stack

Given the mobile/backend nature of this project:

Mobile

Flutter

Advantages:

- Android + iOS
- Strong offline architecture
- Native GPS
- Background capabilities
- Good enterprise UI performance
- One codebase

Potential architecture:

```
Flutter
├── Riverpod / Bloc
├── GoRouter
├── Drift / SQLite
├── Dio
├── Firebase Messaging
├── Google Maps
└── Secure Storage
```

---

135. Backend

A practical stack:

Option A

```
Backend:
NestJS
PostgreSQL
Redis
S3-compatible object storage
RabbitMQ / Kafka where needed
```

Option B

```
ASP.NET Core
PostgreSQL
Redis
Azure Blob
Azure Service Bus
```

For an enterprise pharmaceutical system, .NET is particularly attractive if the target customers are heavily Microsoft-oriented.

---

136. Admin Portal

Possible:

```
React
+
TypeScript
+
Vite
+
TanStack Query
+
MUI / Ant Design
```

---

137. Infrastructure

Potential AWS architecture:

```
CloudFront
    ↓
Load Balancer
    ↓
API
    ↓
ECS / Kubernetes
    ↓
PostgreSQL
    ↓
Redis
    ↓
S3
```

Or Azure:

```
Front Door
 ↓
App Service / Container Apps
 ↓
Azure PostgreSQL
 ↓
Redis
 ↓
Blob Storage
```

Do not choose Kubernetes solely because it sounds enterprise. It adds operational complexity.

---

138. Testing Strategy

Testing must occur at multiple levels.

Unit

Business rules.

Integration

API + database.

Mobile

Offline sync.

UI

User journeys.

Security

Penetration testing.

Performance

Load testing.

Regulatory

Audit and electronic-signature tests where applicable.

---

139. Critical Test Scenarios

Offline

Turn off internet during call.

Expected:

Call saved locally and later synchronized.

Crash

Force-close app after collecting sample signature.

Expected:

Transaction remains recoverable.

Duplicate submission

Tap Submit twice.

Expected:

One call only.

GPS

GPS unavailable.

Expected:

Configurable offline/GPS exception workflow.

Conflict

Two users update customer.

Expected:

Conflict resolution.

Expired sample license

Expected:

Sampling blocked or escalated according to configured policy.

Expired consent

Expected:

Communication blocked.

Unauthorized user

Rep attempts to access another territory.

Expected:

Server rejects request.

---

140. Acceptance Criteria for Tour Plan

A tour-plan implementation is not complete until:

- User can create a plan.
- User can select a cycle.

- System can display target customers.
- User can schedule customers.
- System detects conflicts.
- User can reorder visits.
- Route distance is shown.
- User can submit.
- Manager receives notification.
- Manager can approve/reject.
- Rejected plan returns with reason.
- Approved plan becomes available offline.
- Changes after approval are audited.

---

141. Acceptance Criteria for Visit

A visit implementation is not complete until:

- User can start planned visit.
- User can start unplanned visit.
- System records time.
- System records location.
- System validates configurable GPS rules.
- User sees pre-call information.
- User can record discussion.
- User can add products.
- User can create follow-up.
- User can record samples where enabled.
- User can submit offline.
- Submitted call becomes immutable or amendment-controlled.
- Synchronization is reliable.

---

142. Acceptance Criteria for Offline

The application must be tested under:

- 5G
- 4G
- 3G
- Poor network
- No network
- Intermittent network
- Airplane mode
- App backgrounded
- Device restart

- Battery shutdown

The user should never have to wonder:

> “Did my call disappear?”

---

143. Requirement-Gathering Process

This is how I recommend you actually gather requirements before writing production code.

Phase 1: Stakeholder interviews

Interview:

Field

- Medical reps
- Sales reps
- Area managers
- Regional managers

Corporate

- Sales operations
- Marketing
- Medical affairs
- HR
- Finance
- IT
- Compliance
- Pharmacovigilance

---

144. Questions for Sales Representatives

Ask:

1. How do you receive your daily plan?
2. Who prepares your tour plan?
3. How far in advance?
4. Who approves it?
5. What happens when a doctor isn't available?
6. What constitutes a completed visit?
7. What data must be captured?

8. What samples do you carry?
9. How are samples reconciled?
10. How do you handle unplanned calls?
11. How do you report competitor activity?
12. What happens when the internet disappears?
13. What information do you want before seeing a doctor?
14. What information is painful to enter?
15. What do managers usually ask you for?

These questions frequently expose requirements that executives do not know exist.

---

145. Questions for Managers

Ask:

- How do you assign territories?
- How do you evaluate representatives?
- Which KPIs matter?
- How do you approve tour plans?
- What makes a bad tour plan?
- How do you identify missed doctors?
- How do you perform joint work?
- How do you coach?
- What reports do you manually create?
- Which Excel files do you maintain?

A useful rule:

> Every recurring Excel sheet is potentially an undeveloped software module.

---

146. Questions for Marketing

Ask:

- How are targets selected?
- Who defines segmentation?
- Who defines product messaging?
- Who approves content?
- How are campaigns measured?
- How are doctors targeted?
- How do field representatives receive new content?

---

147. Questions for Medical Affairs

Ask:

- What counts as a medical inquiry?
- What is a medical insight?
- Which users can see it?
- How is it escalated?
- What content is medical-approved?
- How are KOL relationships managed?

---

148. Questions for Pharmacovigilance

Ask:

- What information constitutes an adverse event?
- What must be captured immediately?
- Who receives the report?
- What is the deadline?
- Which integrations exist?
- What records need audit trails?
- Can a rep edit a submitted report?
- How is case identification generated?

This conversation should occur before implementing the AE module.

---

149. Questions for Compliance

Ask:

- Which promotional activities are allowed?
- What samples require signatures?
- How long is consent valid?
- Which content can be used?
- What events require approval?
- What expenses require approval?
- What must be audited?
- What must be retained?
- Who can access what?

---

150. Questions for IT

Ask:

- Existing ERP?
- Existing HRMS?
- Existing CRM?
- Existing HCP provider?
- Identity provider?
- Cloud preference?
- SSO?
- MDM?
- Device-management system?
- Data residency requirements?
- Backup requirements?

---

151. Requirement Workshop Output

Every workshop should produce:

```
Business Process
  ↓
Actors
  ↓
Inputs
  ↓
Steps
  ↓
Business Rules
  ↓
Outputs
  ↓
Exceptions
  ↓
Audit requirements
  ↓
Integrations
  ↓
KPIs
```

This is much better than simply asking:

> “What screens do you want?”

152. Process Mapping Standard

For every major business process document:

Trigger

What starts it?

Actor

Who performs it?

Preconditions

What must already be true?

Main flow

Normal steps.

Alternate flows

Exceptions.

Business rules

What determines success/failure?

Data

What is created or updated?

Approval

Who approves?

Notification

Who gets informed?

Audit

What needs to be recorded?

153. Example Process Requirement

Process

Monthly Tour Planning

Trigger

New cycle begins.

Actor

Representative.

Preconditions

- Employee active.
- Territory assigned.
- Cycle active.
- Targets available.

Main flow

1. Representative opens planning.
2. Selects cycle.
3. Reviews targets.
4. System suggests schedule.
5. Representative modifies schedule.
6. System validates.
7. Representative submits.
8. Manager notified.
9. Manager reviews.
10. Manager approves.

Exceptions

- HCP unavailable
- Leave
- Conflict
- Excessive travel
- Territory mismatch

Output

Approved tour plan.

Audit

All plan edits retained.

---

154. Product Architecture Principle

Organize the product around **business capabilities**, not screens.

Bad architecture:

```
DoctorScreen
DoctorEditScreen
DoctorListScreen
```

Better:

```
CustomerDomain
 ├── Customer
 ├── CustomerAffiliation
 ├── CustomerSegmentation
 ├── CustomerConsent
 ├── CustomerInteraction
 └── CustomerInsight
```

The UI then consumes the business domain.

---

155. Data Ownership

Every important entity should have an owner.

Example:

Data	Owner
Employee	HR
Role	IT/Admin
Territory	Sales Ops
HCP master	Master Data

Data	Owner
Product	Marketing/Product
Sample inventory	Supply/Sample Admin
Content	Medical/Marketing
Consent	Compliance/CRM
Medical inquiry	Medical Affairs
Adverse event	Pharmacovigilance
Expense	Finance
Call report	Field user

---

156. Critical Architectural Decision

Do **not** make the representative's mobile app the system of record.

The mobile app is a field execution client.

The backend is the authoritative system.

```
Mobile
    ↓
Local operational cache
    ↓
Sync
    ↓
Authoritative backend
```

This makes offline operation manageable and protects data consistency.

---

157. The Complete Business Flow

The platform's ultimate lifecycle should look like:

```
ORGANIZATION
     ↓
EMPLOYEES
```

```
      ↓
TERRITORIES
      ↓
CUSTOMERS
      ↓
PRODUCTS
      ↓
SEGMENTATION
      ↓
TARGETING
      ↓
CYCLE PLAN
      ↓
TOUR PLAN
      ↓
DAILY PLAN
      ↓
FIELD VISIT
      ↓
CALL REPORT
      ↓
CONTENT / PRODUCT
      ↓
SAMPLE / ORDER
      ↓
INSIGHT / RESPONSE
      ↓
FOLLOW-UP
      ↓
ANALYTICS
      ↓
NEXT ACTION
      ↓
NEW PLANNING CYCLE
```

This loop is the heart of the application.

---

158. What Will Make This Product Actually Competitive

Do not attempt to beat Veeva by merely copying screens.

The opportunity is to make the product:

Easier

A representative should complete a call report in seconds, not treat it as paperwork.

Faster

Offline-first and optimized data entry.

More configurable

Admin users should change workflows without redeploying the mobile app.

More intelligent

AI should eventually reduce planning and reporting work.

More transparent

Managers should understand not just activity, but outcomes and exceptions.

More locally appropriate

For Pakistan and similar markets, support local operational realities:

- Distributor/stockist workflows
- Territory structures
- Local HCP data
- Local expense practices
- DRAP compliance
- Local geography
- Intermittent connectivity
- Urdu/localization

---

159. Recommended Development Order

Phase 0: Discovery

Do not code yet.

Create:

- Business process maps
- User personas
- Role matrix

- Data dictionary
- Requirements catalogue
- Approval matrix
- Regulatory matrix
- Integration inventory

Phase 1: Foundation

Build:

- Authentication
- Organization
- Employees
- Roles
- Permissions
- Territories
- Master data
- Audit
- Notifications

Phase 2: Field Execution

Build:

- HCP
- HCO
- Product
- Cycle
- Tour plan
- Daily work
- Attendance
- GPS
- Visit
- Call reporting
- Offline sync

This is the first commercially meaningful release.

Phase 3: Commercial Operations

Add:

- Samples

• Inventory
• Orders
• Expenses
• Content
• Events

Phase 4: Medical & Compliance

Add:

• Consent
• Medical inquiries
• Medical insights
• Adverse events
• KOL management
• Compliance workflows

Phase 5: Intelligence

Add:

• Route optimization
• AI summaries
• Predictive analytics
• Next-best action
• AI tour planning
• Voice data entry

---

160. Final Product Blueprint

At maturity, the platform should look conceptually like this:

```
                       ┌─────────────────┐
                       │   ADMIN PORTAL  │
                       │ HR / SALES / IT │
                       └────────┬────────┘
                                │
                                ▼
┌──────────────┐          ┌───────────────────┐
│ MOBILE USERS │          │   CORE PLATFORM   │
│              │◄────────►│                   │
```

```
│ Sales Reps     │     Sync     │ Organization      │
│ MRs            │              │ Customer Master   │
│ Managers       │              │ Territory         │
│ MSLs           │              │ Planning          │
└────────────────┘              │ Field Execution   │
                                │ Samples           │
                                │ Content           │
                                │ Consent           │
                                │ Medical           │
                                │ Compliance        │
                                │ Analytics         │
                                └───────────────────┘
                                         │
                                         │
             ┌───────────────────────────┼───────────────────────┐
             │                           │                       │
             ▼                           ▼                       ▼
            ERP                        HRMS                     MDM
             │                           │                       │
             └───────────────────────────┼───────────────────────┘
                                         │
                                         ▼
                                 DATA / ANALYTICS
                                         │
                                         ▼
                                        AI
                                         │
                ┌────────────────────────┼──────────────────┐
                ▼                        ▼                  ▼
            Next Best              Forecasting         AI Planning
             Action
```

---

161. Final Recommendation

The most important architectural decision is to treat this as **three connected products**:

Product 1: Field Force Mobile

The representative's execution environment.

Its job is:

> Plan → Navigate → Engage → Record → Complete.

Product 2: Management & Administration Portal

Management's control environment.

Its job is:

> Configure → Assign → Approve → Monitor → Coach → Analyze.

Product 3: Life Sciences Backend Platform

The organizational brain.

Its job is:

> Store → Validate → Synchronize → Govern → Integrate → Audit → Analyze.

Build those three layers around a common domain model.

The competitive analysis shows that the market has moved considerably beyond traditional “Sales Force Effectiveness” software. Veeva, Salesforce, IQVIA and Exeevo are converging around unified customer data, territory and account planning, field execution, omnichannel engagement, samples, consent, offline capability, analytics and AI.

For a new product, therefore, the correct foundational chain is:

Organization → Territory → HCP/HCO Master → Product → Segmentation → Cycle → Tour Plan → Daily Work → Visit → Call Report → Samples/Orders/Insights → Follow-up → Analytics → Next Best Action.

That chain should become the backbone of the database, API, mobile application, admin portal, requirements catalogue and test strategy.

Finally, treat security and regulatory requirements as architecture requirements from day one. OWASP MASVS provides a mobile-security baseline, ISO/IEC 27001 provides an information-security management framework, and pharma deployments may also invoke requirements around electronic records/signatures, consent, sampling, pharmacovigilance and local regulatory obligations.

The next engineering artifact I would derive from this specification is a **complete system blueprint containing the ERD/database schema, API specification, screen-by-screen mobile UX, web-admin sitemap, RBAC matrix, state machines, and the exact MVP backlog broken into epics → features → user stories → acceptance criteria**. That is the level at which this requirements document becomes directly executable by a development team.